

Software Requirements Specification

Heart Disease Prediction System

Document Version: 1.0

Prepared for: Academic Submission

Classification: Project Documentation | 2025

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the complete functional and non-functional requirements for the Heart Disease Prediction System. It is intended for developers, testers, evaluators, and college faculty.

1.2 Scope

The Heart Disease Prediction System is a web-based application that enables users to enter clinical patient data and receive an instant prediction of the likelihood of heart disease. The system employs a supervised machine learning model trained on the UCI Heart Disease Dataset.

1.3 Objectives

- Provide an accurate ($\geq 85\%$) machine-learning-based heart disease prediction
- Offer a clean, intuitive web interface accessible from any modern browser
- Validate all user inputs before passing them to the model
- Display prediction results with confidence scores
- Support easy deployment on local machines and cloud platforms

1.4 Definitions and Abbreviations

Term	Definition
SRS	Software Requirements Specification
ML	Machine Learning
UI	User Interface

Term	Definition
RF	Random Forest (classification algorithm)
UCI	University of California Irvine (dataset source)
pkl	Pickle / Joblib serialised model file

2. Overall Description

2.1 Product Perspective

The system is a standalone web application built using Python and Streamlit. It does not require a database or external API. The machine learning model is pre-trained and stored as a binary file (model.pkl) loaded at runtime.

2.2 User Classes

User Class	Description	Interaction
Patient / General User	Non-technical individual seeking health information	Enters data, views result
Medical Practitioner	Doctor or nurse using the tool for quick screening	Enters clinical data, interprets result
Developer	Person maintaining or extending the system	Full system access
College Evaluator	Faculty reviewing the project	Demonstration and review

2.3 Assumptions and Dependencies

- Python 3.10+ is installed on the host machine
- The dataset used for training is the UCI Heart Disease (Cleveland) dataset
- Internet access is not required once the app is running locally
- The trained model (model.pkl) is present in the project root
- Users have a modern web browser (Chrome, Firefox, Edge)

3. Functional Requirements

ID	Requirement	Priority
FR-01	The system shall display a web form for entering 13 patient attributes	High
FR-02	The system shall validate each input field for type and range correctness	High
FR-03	The system shall load the pre-trained model on application startup	High
FR-04	The system shall pass validated inputs to the model and obtain a prediction	High
FR-05	The system shall display the prediction (Disease / No Disease) prominently	High
FR-06	The system shall display the prediction probability / confidence score	Medium
FR-07	The system shall show a descriptive error message for invalid inputs	High
FR-08	The system shall allow the user to reset the form and enter new data	Medium
FR-09	The system shall be accessible via a web browser on port 8501	High
FR-10	The system shall support re-training via a separate train_model.py script	Low

4. Non-Functional Requirements

Category	Requirement
Performance	Prediction response time must be less than 2 seconds
Usability	A first-time user should be able to make a prediction within 3 minutes
Reliability	The application must handle unexpected inputs without crashing
Portability	Must run on Windows 10/11, macOS 12+, and Ubuntu 22.04
Maintainability	Code must be modular, PEP-8 compliant, and well-commented
Scalability	Architecture must support adding new ML models without restructuring
Security	No patient data is stored; all processing is stateless

5. Hardware Requirements

Component	Minimum	Recommended
Processor	Intel Core i3 / AMD Ryzen 3	Intel Core i5 or above
RAM	4 GB	8 GB or more
Storage	500 MB free space	1 GB free space
Network	Not required (local)	For cloud deployment
Display	1024 × 768	1920 × 1080

6. Software Requirements

Software	Version	Purpose
Operating System	Windows 10/11 / Ubuntu 22.04 / macOS 12+	Host environment
Python	3.10 or above	Core programming language
Streamlit	1.30+	Web application framework
Scikit-learn	1.4+	Machine learning library
Pandas	2.x	Data manipulation
NumPy	1.x	Numerical computing
Joblib	1.3+	Model serialisation
Matplotlib	3.8+	Data visualisation

Software	Version	Purpose
VS Code / PyCharm	Latest	Development IDE (optional)
Git	2.x	Version control (optional)